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Path properties of kernel generated two-time parameter Gaussian processes

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Summary. We study path properties of two-parameter Gaussian processes $\{X(t, v), t \in \mathbb{R}, v \in \mathbb{R}^+\}$ of the form $X(t, v) = \int_0^\infty \int_{-\infty}^\infty \Gamma(t, v, x, y)\, \mathrm{d}W(x, y)$, where the kernel function $\Gamma(t, v, x, y)$ is assumed to be square integrable in (x, y) on $\mathbb{R} \times \mathbb{R}^+$, and $W(x, y)$ is a standard two-parameter Wiener process.

1. Introduction

This exposition was inspired by some recent studies of infinite dimensional stochastic evolution equations which, in turn, were initiated by D.A. Dawson (1972, 1975), where the solution was used to model continuous space-time population processes. In particular, the roots of this paper are related to the infinite dimensional Gaussian process $Y(t) = (X_1(t), \ldots, X_i(t), \ldots)$, where the $\{X_i(t), -\infty < t < \infty\}$ are independent Ornstein–Uhlenbeck processes with coefficients γ_i and λ_i ($\gamma_i, \lambda_i > 0$), i.e. $EX_i(t) = 0$ and $EX_i(s)X_i(t) = (\gamma_i/\lambda_i)\exp(-\lambda_i|s-t|)$, $i = 1, 2, \ldots$. In Dawson (1972) the process $Y(\cdot)$ is the stationary solution of the infinite array of stochastic differential equations

$$\mathrm{d}X_i(t) = -\lambda_i X_i(t)\,\mathrm{d}t + (2\gamma_i)^{1/2}\,\mathrm{d}W_i(t) \quad (i = 1, 2, \ldots), \tag{1.1}$$

where $\{W_i(t), -\infty < t < \infty\}$ are independent Wiener processes (cf. also Dawson 1975; Walsh 1981, 1984 and Antoniadis and Carmona 1987). Such processes have been used as a model for neuronal behaviour (Kallianpur and Wolpert 1984; Walsh 1981); they have proved important in quantum field theory (Carmona 1977; Röckner 1988); and they also arise as fluctuation limits in infinite particle systems (Holley and Stroock 1978). Owing to the variety of applications of these processes,

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they have been looked at from a number of different mathematical angles. For instance, they have been considered as examples of stochastic evolution equations (DaPrato et al. 1987; Kotelenez 1987; Miyahara 1981); or as reversible Markov processes which may be studied by using the associated theory of Dirichlet forms (Schmuland 1988b, 1989); or as solutions to stochastic p.d.e.'s (Walsh 1981, 1984). They have also been studied directly as Gaussian processes (Antoniadis and Carmona 1987; Csáki et al. 1990; Csörgő and Lin 1988, 1989, 1990; Fernique 1989; Iscoe et al. 1990); and it is the latter lines of research which have lead us to our present deliberations which we now outline.

Concerning the infinite dimensional Ornstein–Uhlenbeck process $Y(\cdot)$ as above, a special case of *Théorème* in Fernique (1989) reads as follows: For each $x \in \mathbb{R}^+$, let $K(x) = \{k \in \mathbb{N} : \gamma_k > \lambda_k x\}$ and $\lambda(x) = \sup\{\lambda_k : k \in K(x)\}$. Then $Y(\cdot) \in l^2$ is a.s. continuous if and only if we have $\sum_{k=1}^{\infty}(\gamma_k/\lambda_k) < \infty$ and $\int((\log[\lambda(x)] \vee 0)\,dx < \infty$ as well. Consequently, (cf. Corollary 1 of Fernique 1989), for $Y(\cdot) \in l^2$ to be a.s. continuous, it is sufficient that we have

$$\sum_{k=1}^{\infty} (\gamma_k/\lambda_k)(1 + ((\log \lambda_k) \vee 0)) < \infty \ . \tag{1.2}$$

For moduli of continuity results for $Y(\cdot) \in l^2$ and for its closely related l^2-norm squared process $\chi^2(t) = \| Y(t) \|^2 = \sum_{k=1}^{\infty} X_k^2(t)$, we refer to Csörgő and Lin (1988, 1990), Schmuland (1988a), and Csáki and Csörgő (1990).

Another real valued process which is also closely related to $Y(\cdot) \in l^2$ is the stationary mean zero Gaussian process $X(\cdot)$ defined by

$$\{X(t), -\infty < t < \infty\} = \left\{ \sum_{k=1}^{\infty} X_k(t), -\infty < t < \infty \right\},$$

where the $X_k(\cdot)$ are the Ornstein–Uhlenbeck components of $Y(\cdot)$. Let

$$\begin{aligned} &\{X(t, n), -\infty < t < \infty, n = 1, 2, \ldots\} \\ &= \left\{ \sum_{k=1}^{n} X_k(t), -\infty < t < \infty, n = 1, 2, \ldots \right\}. \end{aligned} \tag{1.3}$$

Csáki et al. (1990) show that $X(t, n) \to X(t)$ uniformly in t over finite intervals with probability one as $n \to \infty$ if for some $\delta > 0$ we have

$$\sum_{k=1}^{\infty} \gamma_k (\log(\lambda_k \vee e))^{1+\delta}/\lambda_k < \infty \ , \tag{1.4}$$

and hence, under the latter condition, $X(\cdot)$ is continuous with probability one. Csáki et al. (1990) also prove exact moduli of continuity results for $X(\cdot)$. The relationship of $X(\cdot)$ and $Y(\cdot) \in l^2$ is illustrated by conditions (1.2) and (1.4). Indeed, the process $X(\cdot)$ is continuous if and only if it satisfies Fernique's necessary and sufficient condition for the continuity of a stationary Gaussian process (cf. Corollary 2.5 of Sect. IV.2 in Jain and Marcus 1978), i.e., if and only if in this case $E|X(t) - X(s)|^2 = \phi^2(|t - s|)$, where $\phi(u)$ is an increasing function in $u > 0$, we have that $\phi(u)/(u(\log(1/u))^{1/2})$ is integrable at zero. Since

$$E\| Y(t) - Y(s)\|^2 = E|X(t) - X(s)|^2 \ ,$$

hence, in general, checking Fernique's necessary and sufficient condition for the a.s. continuity of the real valued, stationary, mean zero Gaussian process $X(\cdot)$ should

be also sufficient for that of the stationary, mean zero Gaussian process $Y(\cdot)$ in l^2. The conditions (1.2) and (1.4) illustrate this point. More importantly along these lines, a referee of our paper has remarked that a more precise analysis and study of the results of Fernique (1989) yield in effect that the following four properties are equivalent:

(i) $$P\{\forall t,\ Y(t) \in l^2\} = 1\,, \quad P\{t \to Y(t) \text{ continuous}\} = 1\,,$$

(ii) $$\sum_{k=1}^{\infty} \gamma_k/\lambda_k < \infty\,, \quad \int \log^+ \lambda(x)\,\mathrm{d}x < \infty\,,$$

(iii) $$\forall t, \quad P\{X(t, n) \to X(t)\} = 1\,, \quad P\{t \to X(t) \text{ continuous}\} = 1\,,$$

(iv) $$P\{X(\cdot, n) \to X(\cdot) \text{ uniformly over every finite interval}\} = 1\,.$$

Path properties of the two-parameter Gaussian process $X(t, n)$ of (1.3) were studied by Csörgő and Lin (1989). Integrating the equations in (1.1) from $-\infty$ to t we obtain

$$X_i(t) = \int_{-\infty}^{t} \exp(-\lambda_i|t-s|)(2\gamma_i)^{1/2}\,\mathrm{d}W_i(s) \quad (i = 1, 2, \ldots)\,, \tag{1.5}$$

and hence we have also

$$X(t, n) = \sum_{k=1}^{n} X_k(t) = \sum_{k=1}^{n} \int_{-\infty}^{t} \exp(-\lambda_k|t-s|)(2\gamma_k)^{1/2}\,\mathrm{d}W_k(s)\,. \tag{1.6}$$

The latter, in turn, has lead us to study, in the same paper, also the two-parameter Gaussian process

$$X(t, v) = \int_0^v \int_{-\infty}^{t} \exp(-\lambda(y)|t-x|)(2\gamma(y))^{1/2}\,\mathrm{d}W(x, y)\,, \tag{1.7}$$

where $\gamma(y)$ and $\lambda(y)$ are assumed to be positive continuous functions on $[0, \infty)$, and $\{W(x, y), -\infty < x < \infty, 0 \leqq y < \infty\}$ is a standard two-parameter Wiener process (cf. Sects. 1.10–1.15 and the Supplementary remarks of Chap. 1 in Csörgő and Révész 1981).

This brings us to the main topic of our present exposition. Here we study two-parameter Gaussian processes $\{X(t, v), t \in \mathbb{R}, v \in \mathbb{R}^+\}$ of the form

$$X(t, v) = \int_0^{\infty} \int_{-\infty}^{\infty} \Gamma(t, v, x, y)\,\mathrm{d}W(x, y)\,, \tag{1.8}$$

where the kernel function $\Gamma(t, v, x, y)$ is assumed to be square integrable in (x, y) on $\mathbb{R} \times \mathbb{R}^+$ and $W(x, y)$ is a standard two-parameter Wiener process. Thus $X(t, v)$ is a Gaussian process with mean zero and covariance function

$$\mathrm{Cov}(X(t, v), X(s, u)) = \int_0^{\infty} \int_{-\infty}^{\infty} \Gamma(t, v, x, y)\Gamma(s, u, x, y)\,\mathrm{d}x\,\mathrm{d}y\,. \tag{1.9}$$

We note that we do not assume stationarity properties for $X(t, v)$. Put

$$k(t, s, v, u, y) = \int_{-\infty}^{\infty} \Gamma(t, v, x, y)\Gamma(s, u, x, y)\,\mathrm{d}x \tag{1.10}$$

and call this function covariance kernel function. The following examples are immediate.

Example 1. If $\Gamma(t, v, x, y) = \mathbf{1}_{(-\infty, t] \times [0, v]}(x, y)$, $0 \leqq v < \infty$, $-\infty < t < \infty$, then $X(t, v) = W(t, v)$ and

$$k(t, s, v, u, y) = \begin{cases} t \wedge s, & 0 \leqq y \leqq v \wedge u\,, \\ 0, & \text{otherwise}\,. \end{cases}$$

Example 2. If $\Gamma(t, v, x, y) = \mathbf{1}_{[0, t] \times [0, v]}(x, y) - t\mathbf{1}_{[0, 1] \times [0, v]}(x, y)$, $0 \leqq t \leqq 1$, $0 \leqq v < \infty$, then $X(t, v) = W(t, v) - tW(1, v)$, a Kiefer process (cf. Sect. 1.15 in Csörgő and Révész 1981) and

$$k(t, s, v, u, y) = \begin{cases} t \wedge s - ts, & 0 \leqq y \leqq v \wedge u\,, \\ 0, & \text{otherwise}\,. \end{cases}$$

Example 3. If, with $-\infty < t < \infty$, $0 \leqq v < \infty$,

$$\Gamma(t, v, x, y) = \mathbf{1}_{(-\infty, t] \times [0, v]}(x, y) \exp(-\lambda(y)(t - x))(2\gamma(y))^{1/2}\,,$$

where $\lambda(y)$ and $\gamma(y)$ are positive continuous functions on $[0, \infty)$, then $X(t, v)$ is the two-parameter Gaussian process of (1.7) and

$$k(t, s, v, u, y) = \begin{cases} (\gamma(y)/\lambda(y)) \exp(-\lambda(y)|t - s|), & 0 \leqq y \leqq v \wedge u\,, \\ 0, & \text{otherwise}\,, \end{cases}$$

is the Ornstein–Uhlenbeck covariance kernel function fashioned after the covariance function of the $X_i(\cdot)$ of (1.5).

We put

$$\Delta_1 k(t, s, v, y) = k(t + s, t + s, v, v, y) - 2k(t, t + s, v, v, y) + k(t, t, v, v, y)\,. \tag{1.11}$$

Then we have

$$E\{X(t + s, v) - X(t, v)\}^2 = \int_0^\infty \Delta_1 k(t, s, v, y)\,\mathrm{d}y := H_1(t, s, v)\,. \tag{1.12}$$

If H_1 does not depend on t, then the process $X(t, v)$ is weakly stationary in its parameter t. If we can write

$$H_1(t, s, v) = \int_t^{t+s} k_1(v, y)\,\mathrm{d}y\,,$$

then, by direct calculations, $X(t, v)$ is a process with independent increments in t. Similarly to (1.12), on putting

$$H_2(t, v, u) = E\{X(t, v + u) - X(t, v)\}^2\,,$$

the same cases of the process $X(t, v)$ can be considered and similarly described for its parameter v. For the sake of studying increments of $X(t, v)$ in both of its parameters t and v we put

$$X(R(t, s, v, u)) = X(t + s, v + u) - X(t, v + u) - X(t + s, v) + X(t, v)\,, \tag{1.13}$$

and get

$$E\{X(R(t, s, v, u))\}^2 = \int_0^\infty \Delta_2 k(t, s, v, u, y)\,\mathrm{d}y := H_3(t, s, v, u)\,, \tag{1.14}$$

where

$$\begin{aligned}\Delta_2 k(t, s, v, u, y) = {} & \Delta_1 k(t, s, v, y) + \Delta_1 k(t, s, v + u, y) \\ & - 2k(t + s, t + s, v, v + u, y) + 2k(t, t + s, v, v + u, y) \\ & + 2k(t, t + s, v + u, v, y) - 2k(t, t, v, v + u, y) .\end{aligned}$$

The motivation of this paper is to study the problem of generating further two-time parameter Gaussian processes out of a Wiener sheet. As illustrated by Examples 1, 2 and 3 above, it is the kernel function Γ of (1.8) that has the role of generating these two-time parameter Gaussian processes of interest. The main aim of this exposition is to establish moduli of continuity forms for the general two-parameter Gaussian process $X(t, v)$ of (1.8) as $v \to \infty$ (cf. Sect. 3). These continuity moduli are based on large deviations for appropriate increments of $X(t, v)$. Inevitably then, our conditions on the kernel function Γ in these large deviation theorems (cf. Sect. 2) are quite complicated in their general appearance. They are verified for the Kiefer process of Example 2 as well as for the process of (1.7), which is discussed in Example 3, in our Remarks 2.2–2.6, where other examples are also reviewed briefly. In particular, in Remark 2.2 we give stronger but simpler conditions in order to explain the meaning of the more general ones for weakly stationary processes. The Wiener sheet $W(t, v)$ of Example 1 is well known. A verification of our present general conditions for $W(t, v)$ is immediate along the lines of Remark 2.3, concerning the Kiefer process.

This paper is organized as follows. In Sect. 2 we state and comment on our large deviation results for some increments of the process $X(t, v)$ as in (1.8). We use these results to prove our main theorems concerning path properties of the latter process, which themselves are stated and commented on in Sect. 3. In Sect. 4 we prove the lemmata of Sect. 2, which are also of interest on their own, and in Sect. 5 we prove the results of Sect. 3.

2. Large deviations

In the sequel we denote constants depending on ε by $C = C(\varepsilon)$, which may take different values at different places. Moreover, in order to simplify the statements of our lemmata, we make use of the following convention: the inequalities here are meant to hold true for large V and small $\varepsilon > 0$.

Let $e(j)$ and $g(j)$ be two positive functions, tending to infinity monotonically, and let $d_V = d(V)$ be a positive continuous function. Put $r' = r'(r) = (d_V e(r))^{-1}$. For $t \in \mathbb{R}$ and integer r define $t_r = [t/r']r'$. Let $\varepsilon > 0$, $u_V > 0$, $s_V > 0$ and $0 < T_V \uparrow \infty$ $(V \to \infty)$.

Lemma 2.1. *Suppose that for given $\delta > 0$ there exists $\delta_1 > 0$ such that for $|t| \leqq T_V$, $t_r \leqq t_1 \leqq t$, $0 \leqq s_1 \leqq s_V(1 + \delta_1)$ and large r we have*

$$H_1(t_1, s_1, V)/H_1(t, s_V, V) \leqq 1 + \delta . \tag{2.1}$$

Furthermore, suppose that

$$\liminf_{V \to \infty} d_V s_V > 0 , \tag{2.2}$$

$$\sum_{j=0}^{\infty} e^2(r + j + 1) \exp(-g(j)/(2 + \varepsilon)\} < \infty \text{ for every } r > 0 \tag{2.3}$$

and for given $\delta > 0$ we have also

$$\sum_{j=0}^{\infty} (g(j)H_1((t+s)_{r+j}, r'(r+j+1), V))^{1/2} \leqq \delta(H_1(t, s_V, V))^{1/2} \tag{2.4}$$

for $|t| \leqq T_V$, $0 \leqq s \leqq s_V$, if r is large enough. Then

$$P\left\{\sup_{|t| \leqq T_V} \sup_{0 \leqq s \leqq s_V} (H_1(t, s_V, V))^{-1/2} |X(t+s, V) - X(t, V)| \geqq x\right\} \leqq C d_V^2 T_V s_V \exp\{-x^2/(2+\varepsilon)\}. \tag{2.5}$$

If, in addition, for $V_1 \leqq v \leqq V$, $|t| \leqq T_V + s_V$, $0 \leqq s \leqq s_V + r'(r)$, $0 \leqq u_1$, $u_2 \leqq V - V_1$, satisfying $0 \leqq (V - V_1)/V \to 0$ as $V \to \infty$, we have

$$\lim_{V \to \infty} H_1(t, s, v)/H_1(t, s, V) = 1 \tag{2.6}$$

and

$$H_3(t, s, v, u_1) + H_3(t, s, v + u_1, u_2) \leqq H_3(t, s, v, u_1 + u_2), \tag{2.7}$$

then

$$P\left\{\sup_{V_1 \leqq v \leqq V} \sup_{|t| \leqq T_V} \sup_{0 \leqq s \leqq s_V} (H_1(t, s_v, v))^{-1/2} |X(t+s, v) - X(t, v)| \geqq x\right\} \leqq C d_V^2 T_V s_V \exp\left\{-\frac{x^2}{2+\varepsilon}\right\}. \tag{2.8}$$

Lemma 2.2. *Suppose that for given $\delta > 0$ there exist $\delta_1, \delta_2 > 0$ such that*

$$H_3(t_1, s_1, v_1, u_1)/H_3(t, s, v, u) \leqq 1 + \delta \tag{2.9}$$

and

$$H_3(t, s, v, \delta_2 u)/H_3(t, s, v, u) \leqq \delta \tag{2.10}$$

for $|t| \leqq T_V$, $t_r \leqq t_1 \leqq t$, $0 \leqq s_1 \leqq s(1+\delta_1)$, $s \leqq s_V$, $v^r \leqq v_1 \leqq v$, $0 \leqq u_1 \leqq u(1+\delta_1)$, $u \leqq u_v$ and large r. Assume that (2.2), (2.3) and (2.7) hold true and that for any $\delta > 0$

$$\sum_{j=0}^{\infty} (g(j)H_3(t, r'(r+j+1), v, u))^{1/2} \leqq \delta(H_3(t, s_V, v, u))^{1/2} \tag{2.11}$$

for $-T_V \leqq t \leqq T_V + s_V$, $0 \leqq v \leqq V$, $0 \leqq u \leqq u_V$ if r is large enough. Then for $X(R(t, s, v, u))$ as in (1.13) we have

$$P\left\{\sup_{0 \leqq v \leqq V} \sup_{0 \leqq u \leqq u_V} \sup_{|t| \leqq T_V} \sup_{0 \leqq s \leqq s_V} (H_3(t, s_V, v, u_V))^{-1/2} |X(R(t, s, v, u))| \geqq x\right\} \leqq \frac{C d_V V T_V s_V}{u_V} \exp\left\{-\frac{x^2}{2+\varepsilon}\right\}. \tag{2.12}$$

Let $e'(j)$, $g'(j)$ and d'_V be functions like $e(j)$, $g(j)$ and d_V respectively, which are defined in the paragraph just preceding Lemma 2.1. Put $r'' = r''(r) = (d'_V e'(r))^{-1}$ and $v^r = [v/r'']r''$.

Lemma 2.3. *Suppose that for a given $\delta > 0$ we have*

$$H_1(t_1, s_1, v^r)/H_1(t, s_V, v) \leqq 1 + \delta \tag{2.13}$$

for $|t| \leqq T_V$, $t_r \leqq t_1 \leqq t + s_V$, $0 \leqq s_1 \leqq s_V + r'(r)$, $0 \leqq v \leqq V$ *and large* r. *Assume also that* (2.2) *and* (2.3) *are satisfied and for given* $\delta > 0$

$$\sum_{j=0}^{\infty} (g(j)H_1((t+s)_{r+j}, r'(r+j+1), v^r))^{1/2} \leqq \delta(H_1(t, s_V, v))^{1/2} \tag{2.14}$$

for $|t| \leqq T_V$, $0 \leqq s \leqq s_V$, $0 \leqq v \leqq V$, *provided* r *is large enough. Furthermore assume that*

$$\liminf_{V \to \infty} d'_V u_V > 0 , \tag{2.15}$$

$$\sum_{j=0}^{\infty} e'(r+j+1) \exp\left\{ - \frac{g'(j)}{2+\varepsilon} \right\} < \infty , \tag{2.16}$$

and for given $\delta > 0$ *we have*

$$\sum_{j=0}^{\infty} (g'(j)H_2((t+s)_r, v^{r+j}, r''(r+j+1)))^{1/2} \leqq \delta(H_1(t, s_V, v))^{1/2} , \tag{2.17}$$

$$\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} (g(i)g'(j)H_3((t+s)_{r+i}, r'(r+i+1), v^{r+j}, r''(r+j+1)))^{1/2} \leqq \delta(H_1(t, s_V, v))^{1/2} \tag{2.18}$$

for any $|t| \leqq T_V$, $0 \leqq s \leqq s_V$, $0 \leqq v \leqq V$, *provided* r *is large enough. Then*

$$P\left\{ \sup_{0 \leqq v \leqq V} \sup_{|t| \leqq T_V} \sup_{0 \leqq s \leqq s_V} (H_1(t, s_V, v))^{-1/2} |X(t+s, v) - X(t, v)| \geqq x \right\} \leqq C d'_V d_V^2 V T_V s_V \exp\left\{ - \frac{x^2}{2+\varepsilon} \right\} . \tag{2.19}$$

Another version of Lemma 2.3 considers increments of $X(t, v)$ in v.

Lemma 2.4. *Suppose that for a given* $\delta > 0$ *we have*

$$H_2(t_1, v_1, u_1)/H_2(t, v, u_V) \leqq 1 + \delta \tag{2.20}$$

for $|t| \leqq T_V$, $t_r \leqq t_1 \leqq t$, $0 \leqq v \leqq V$, $v^r \leqq v_1 \leqq v + u_V$, $0 \leqq u_1 \leqq u_V + r''(r)$ *and large* r. *Assume also that* (2.2), (2.3), (2.15) *and* (2.16) *are satisfied. Furthermore suppose that for given* $\delta > 0$ *and any* $|t| \leqq T_V$, $0 \leqq v \leqq V$, $0 \leqq u \leqq u_V$,

$$\sum_{j=0}^{\infty} (g'(j)H_2(t_r, (v+u)^{r+j}, r''(r+j+1)))^{1/2} \leqq \delta(H_2(t, v, u_V))^{1/2} , \tag{2.21}$$

$$\sum_{j=0}^{\infty} (g(j)H_1(t_{r+j}, r'(r+j+1), (v+u)^r))^{1/2} \leqq \delta(H_2(t, v, u_V))^{1/2} , \tag{2.22}$$

$$\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} (g(i)g'(j)H_3(t_{r+i}, r'(r+i+1), (v+u)^{r+j}, r''(r+j+1)))^{1/2} \leqq \delta(H_2(t, v, H_V))^{1/2} , \tag{2.23}$$

provided r *is large enough. Then*

$$P\left\{ \sup_{0 \leqq v \leqq V} \sup_{0 \leqq u \leqq u_V} \sup_{|t| \leqq T_V} (H_2(t, v, u_V))^{-1/2} |X(t, v+u) - X(t, v)| \geqq x \right\} \leqq C d_V d_V'^2 V u_V T_V \exp\left\{ - \frac{x^2}{2+\varepsilon} \right\} . \tag{2.24}$$

Remark 2.1. We can similarly consider T, s_T, V_T, u_T instead of T_V, s_V, V, u_V respectively, and obtain inequalities corresponding to (2.5), (2.8), (2.12), (2.19), (2.24).

Remark 2.2. If $X(t, v)$ is weakly stationary in t and $H_1(t, s, v)$ approximately is directly proportional to s^α for some $\alpha > 0$, i.e. there exists some $H(v)$ such that $H_1(t, s, V)/s^\alpha H(V) \to 1$ as $V \to \infty$, then we can take $e(r) = 2^r$, $g(j) = 6j$, $d_V = s_V^{-1}$ and see easily that conditions (2.1)–(2.4) are satisfied. Hence (2.5) holds true. Consequently, for the process $X(t, v)$ of (1.7) (cf. Example 3), the continuous parameter version of (1.1) of Lemma 1.3 in Csörgő and Lin (1989) is a special case of (2.5) here, since for $0 \leqq s \leqq s_V$ we have

$$H_1(t, s, v) = 2 \int_0^v \frac{\gamma(x)}{\lambda(x)} (1 - \mathrm{e}^{-s\lambda(x)})\, \mathrm{d}x \sim 2s \int_0^v \gamma(x)\, \mathrm{d}x \tag{2.25}$$

under the assumption of $(\max_{0 \leqq x \leqq v} \lambda(x)) s_v \to 0$ as $v \to \infty$. In case the latter assumption is not satisfied, we take $d_V = \max_{1 \leqq x \leqq V} \lambda(x)$ and get

$$\begin{aligned} \sum_{j=0}^{\infty} (g(j) H_1((t+s)_{r+j}, r'(r+j+1), V))^{1/2} &\leqq \sum_{j=0}^{\infty} \left(6j 2^{-(r+j)} \int_0^V \frac{\gamma(v)}{\lambda(x)}\, \mathrm{d}x \right)^{1/2} \\ &\leqq \delta \left(\int_0^V \frac{\gamma(x)}{\lambda(x)}\, \mathrm{d}x \right)^{1/2}, \end{aligned}$$

provided that r is large enough. Consequently (2.5) also implies a continuous version of (1.4) of Lemma 1.3 in Csörgő and Lin (1989).

Remark 2.3. If $\{X(t, v),\ 0 \leqq t \leqq 1,\ 0 \leqq v < \infty\}$ is a Kiefer process then $H_1(t, s, v) = v(s - s^2)$. Taking again $e(r) = 2^r$, $g(j) = 6j$, $d_V = s_V^{-1}$, we get (2.5) with Vs_V instead of $H_1(t, s_V, V) = V(s_V - s_V^2)$ if $s_V \to 0$ as $V \to \infty$.

Remark 2.4. For the process $X(t, v)$ of (1.7) (cf. Example 3)

$$H_3(t, s, v, u) = 2 \int_v^{v+u} \frac{\gamma(x)}{\lambda(x)} (1 - \mathrm{e}^{-s\lambda(x)})\, \mathrm{d}x\,. \tag{2.26}$$

If $(\max_{0 \leqq x \leqq v+u_v} \lambda(x)) s_v \to 0$ as $v \to \infty$, and if for any $\delta > 0$ there exists $\delta_1 > 0$ such that $\int_v^{v+\delta_1 u} \gamma(x)\, \mathrm{d}x / \int_v^{v+u} \gamma(x)\, \mathrm{d}x < \delta$, then the conditions of Lemma 2.2 are satisfied for $e(r) = 2^r$, $g(j) = 6j$, $d_V = s_V^{-1}$. Consequently (2.12) implies a continuous version of (1.3) in Lemma 1.3 of Csörgő and Lin (1989).

Remark 2.5. For a Kiefer process $H_3(t, s, v, u) = u(s - s^2)$. So taking $d_V = s_V^{-1}$, we get (2.12) with $u_V s_V$ instead of $u_V(s_V - s_V^2)$ if $s_V \to 0$ as $V \to \infty$.

Remark 2.6. For Theorem 3.2 in Csörgő and Lin (1989) $H_3(t, s, v, u)$ is defined by (2.26) and $H_2(t, v, u) = 2 \int_v^{v+u} \frac{\gamma(x)}{\lambda(x)}\, \mathrm{d}x$. By taking $g(j) = g'(j) = 6j$, $e(r) = e'(r) = 2^r$, $d_V = \lambda'_V =: \max_{0 \leqq x \leqq V+uv} \lambda(x)$, $d'_V = u_V^{-1}$ then, under assumption (3.12) in Csörgő and Lin (1989) which requires the existence of $b > 0$ such that

$$\int_v^{v+\delta u} \frac{\gamma(x)}{\lambda(x)}\, \mathrm{d}x \Big/ \int_v^{v+u} \frac{\gamma x)}{\lambda(x)}\, \mathrm{d}x \leqq b\delta$$

for any $v \in \mathbb{R}^+$ and all small positive δ and u, the conditions of our Lemma 2.4 hold

true, and from this lemma we obtain

$$P\left\{\sup_{0\leqq v\leqq V}\ \sup_{0\leqq u\leqq u_V}\ \sup_{|t|\leqq T_V}\left(2\int_v^{v+u_V}(\gamma(x)/\lambda(x))\,dx\right)^{-1/2}|X(t,v+u)-X(t,v)|\geqq x\right\}$$
$$\leqq\frac{C\lambda_V' VT_V}{u_V}\exp\left\{-\frac{x^2}{2+\varepsilon}\right\}.$$

Comparing the latter with the inequality stated just preceding (3.18) in Csörgő and Lin (1989), which was obtained there by using an inequality of Fernique (1964), we note that here we have improved somewhat in that we have succeeded in dropping a non-decreasing factor λ_V'.

For obtaining our results on path properties, we will also make use of the classical Slepian inequality.

Lemma 2.5 (Slepian 1962). *Let $G(t)$ and $G^*(t)$ be Gaussian processes on $\mathbb{R}^+$, possessing continuous sample path functions, with $EG(t)=EG^*(t)=0$, $EG^2(t)=EG^{*2}(t)=1$, and let $\rho(s,t)$ and $\rho^*(s,t)$ be their respective covariance functions. Suppose that we have $\rho(s,t)\geqq\rho^*(s,t)$, $s,t\in\mathbb{R}^+$. Then*

$$P\left\{\sup_{0\leqq t\leqq T}G(t)\leqq u\right\}\geqq P\left\{\sup_{0\leqq t\leqq T}G^*(t)\leqq u\right\}.$$

3. Path properties

Applying the large deviation results of the previous section for increments of the Gaussian process $X(t,v)$ as in (1.8) and the classical Slepian inequality, here we state and then comment on a series of path properties of this two-time parameter process.

Theorem 3.1. *Suppose that the conditions leading up to having inequality* (2.8) *are satisfied, and now assume also that $T_V\uparrow\infty$, $s_V\downarrow 0$, $d_V\uparrow\infty$ continuously as $V\to\infty$, as well as*

$$\frac{\log(T_V/s_V)}{\log(d_V^2 s_V^2\log V)}\to\infty\,, \tag{3.1}$$

$$\frac{H_3(t,s_V,V-V^{-A},V^{-A})}{H_1(t,s_V,V)}\to 0 \tag{3.2}$$

for some $A>0$, where $|t|\leqq T_V(1+o(1))$. In addition to (2.1) *suppose that for $|t|\leqq T_V(1+o(1))$ as $V\to\infty$ and for $t\leqq t'\leqq t+s_V$ we have also*

$$\frac{H_1(t',\delta_1 s_V,V)}{H_1(t,s_V,V)}\leqq\delta \tag{3.3}$$

if d_V does not depend on s_V, while otherwise assume

$$\frac{H_1(t',\delta s_V,V)}{H_1(t,s_V,V)}\leqq b\delta \tag{3.4}$$

for some $b>0$. Furthermore assume that condition (2.11) *is satisfied and that the*

increments

$$\{X((k+1)s_V, V) - X(ks_V, V), |k| \leqq [T_V/s_V]\}$$

are not positively correlated. Then

$$\text{(3.5)} \quad \lim_{V\to\infty} \sup_{|t|\leqq T_V} (2H_1(t, s_V, V)\log(T_V/s_V))^{-1/2}|X(t+s_V, V) - X(t, V)| = 1 \quad \text{a.s.},$$

$$\text{(3.6)} \quad \lim_{V\to\infty} \sup_{|t|\leqq T_V} \sup_{0\leqq s\leqq s_V} (2H_1(t, s_V, V)\log(T_V/s_V))^{-1/2}|X(t+s, V) - X(t, V)| = 1 \quad \text{a.s.}$$

Theorem 3.2. *Suppose that the conditions leading up to having inequality* (2.12) *are satisfied, and now assume also that* $T_V \uparrow \infty$, $s_V \downarrow 0$, $u_V \downarrow 0$ (*or* u_V *is increasing and* Vu_V^{-1} *is non-decreasing*), $d_V \uparrow \infty$ *continuously as* $V \to \infty$ *and, in addition to* (2.10),

$$\text{(3.7)} \quad \frac{H_3(t, s_V, v+u_V, \delta_2 u_V)}{H_3(t, s_V, v, u_V)} \leqq \delta .$$

Furthermore assume that

$$\text{(3.8)} \quad \frac{H_3(t, \delta_2 s_V, v, u_V)}{H_3(t, s_V, v, u_V)} \leqq \delta$$

if d_V *does not depend on* s_V, *while otherwise assume that for some* $b > 0$ *we have*

$$\text{(3.9)} \quad \frac{H_3(t, \delta s_V, v, u)}{H_3(t, s_V, v, u)} \leqq b\delta .$$

Suppose also that as $V \to \infty$ *we have*

$$\text{(3.10)} \quad \frac{\log(VT_V s_V^{-1} u_V^{-1})}{\log(d_V^2 s_V^2)} \to \infty$$

and that the increments (*cf.* (1.13))

$$\{X(R(ks_V, s_V, lu_V, u_V)), 0 \leqq k \leqq [T_V/s_V]\}$$

and

$$\{X(R(ks_V, s_V, lu_V, u_V)), 0 \leqq l \leqq [V/u_V]\}$$

are both not positively correlated. Then

$$\text{(3.11)} \quad \lim_{V\to\infty} \sup_{0\leqq v\leqq V} \sup_{|t|\leqq T_V} (2H_3(t, s_V, v, u_V)\log(VT_V s_V^{-1} u_V^{-1}))^{-1/2}|X(R(t, s_V, v, u_V))| = 1 \quad \text{a.s.},$$

$$\text{(3.12)} \quad \lim_{V\to\infty} \sup_{0\leqq v\leqq V} \sup_{0\leqq u\leqq u_V} \sup_{|t|\leqq T_V} \sup_{|s|\leqq s_V} (2H_3(t, s_V, v, u_V)\log(VT_V s_V^{-1} u_V^{-1}))^{-1/2} \times |X(R(t, s, v, u))| = 1 \quad \text{a.s.}$$

Remark 3.1. Theorems 3.1 and 3.2 imply the respective continuous parameter versions of Theorems 2.1 and 2.3 in Csörgő and Lin (1989).

Remark 3.2. If $X(t, v)$ is stationary in t, then it can be seen by checking the proofs of Theorems 3.1 and 3.2 that the condition $s_V \downarrow 0$ as $V \to \infty$ can be replaced by the weaker assumptions requiring that $s_V \leqq T_V$, and $T_V s_V^{-1}$ is non-decreasing as

$V \to \infty$. Similarly, if $X(t, v)$ is stationary in v, condition $u_V \downarrow 0$ can be weakened to requiring $u_V \leqq V$, and Vu_V^{-1} is non-decreasing as $V \to \infty$. Consequently, in these cases we may also consider studying the problem of how large the increments of $X(t, v)$ are along the lines of those of a Wiener sheet, for example (cf. Sects. 1.10–1.15 and the supplementary remarks of Chap. 1 in Csörgő and Révész 1981).

Remark 3.3. It will be seen from the proof of Theorem 3.1 that condition (3.3) can be replaced by

$$\sup_{|t| \leqq T_{v_k}, v_{k-1} \leqq v \leqq v_k} \frac{H_1(t, s_{v_{k-1}} - s_{v_k}, v)}{H_1(t, s_{v_k}, v)} \to 0 \tag{3.13}$$

as $k \to \infty$, provided that v_k satisfies (5.2) as well as (5.9) and (5.10). Hence we can for example take v_k such that $T_{v_k} \leqq k^{1/2}$. If s_V is non-decreasing as $V \to \infty$, then $d_V^2 s_V^2 \leqq T_V$ by (3.1). Then, as $k \to \infty$, $d_{v_k}(s_{v_k} - s_{v_{k-1}}) \to 0$, and for Theorem 2.2 in Csörgő and Lin (1989) we have

$$H_1(t, s_{v_k}, v) \sim 2 \int_0^v \frac{\gamma(x)}{\lambda(x)} \, dx \quad \text{and} \quad H_1(t, s_{v_k} - s_{v_{k-1}}, v) \sim o\left(\int_0^v \frac{\gamma(x)}{\lambda(x)} \, dx \right)$$

on account of $d_{v_k}(s_{v_k} - s_{v_{k-1}}) = \lambda'_{v_k}(s_{v_k} - s_{v_{k-1}}) \to 0$ as $k \to \infty$. So (3.13) is satisfied. Under the conditions of Theorem 2.2 in Csörgő and Lin (1989) the further conditions of Theorem 3.1 are satisfied. Hence Theorem 3.1 implies a continuous parameter version of Theorem 2.2 in Csörgő and Lin (1989).

Similarly, Theorem 3.2 implies a continuous parameter version of Theorem 2.4 in Csörgő and Lin (1989).

Theorem 3.3. *Suppose that the conditions leading up to having inequalities* (2.24) *and* (2.12) *are satisfied, and now assume also that* $T_V \uparrow \infty$, $s_V \downarrow 0$, $u_V \downarrow 0$, $d_V \uparrow \infty$, $d'_V \uparrow \infty$ *continuously as* $V \to \infty$, *as well as*

$$\frac{\log(Vu_V^{-1})}{\log(d_V d_V'^2 T_V u_V^2)} \to \infty \,, \tag{3.14}$$

and that for $T_V - s'_V \leqq t \leqq T_V$, $s'_V \downarrow 0$ *as* $V \to \infty$, $0 \leqq v \leqq V$ *we have also*

$$\frac{H_3(t, s'_V, v, u_V)}{H_2(t, v, u_V)} \to 0 \tag{3.15}$$

when $V \to \infty$. *Assume that for any* $\delta > 0$ *there exists* $\delta_1 > 0$ *such that for* $0 \leqq v \leqq V$, $0 \leqq u \leqq u_V$

$$\frac{H_2(T_V, v + u_V, \delta_1 u)}{H_2(T_V, v, u)} \leqq \delta \tag{3.16}$$

if d'_V *does not depend on* u_V, *while otherwise assume that for* $t = T_V(1 + o(1))$ *as* $V \to \infty$ *we have*

$$\frac{H_2(t, v + u_V, \delta u_V)}{H_2(t, v, u_V)} \leqq b\delta \tag{3.17}$$

for some $b > 0$. *Furthermore assume that the increments*

$$\{X(T_V, (k+1)u_V) - X(T_V, ku_V), 0 \leqq k \leqq [V/u_V]\}$$

are not positively correlated. Then

$$\limsup_{V\to\infty} \sup_{0\leqq v\leqq V} (2H_2(T_V, v, u_V)\log(Vu_V^{-1}))^{-1/2}|X(T_V, v+u_V) - X(T_V, v)| = 1 \quad \text{a.s.}, \tag{3.18}$$

$$\lim_{V\to\infty} \sup_{0\leqq v\leqq V} \sup_{0\leqq u\leqq u_V} \sup_{|t|\leqq T_V} (2H_2(t, v, u_V)\log(Vu_V^{-1}))^{-1/2} \times |X(t, v+u) - X(t, v)| = 1 \quad \text{a.s.} \tag{3.19}$$

Remark 3.4. It will be seen from the proof of Theorem 3.3 that condition (3.15) can be replaced by

$$\frac{H_3(t, T_{v_k} - T_{v_{k-1}}, v, u_{v_k})}{H_2(t, v, u_{v_k})} \to 0 \tag{3.20}$$

as $k\to\infty$, for $T_{v_{k-1}} < t \leqq T_{v_k}$, $0 \leqq v \leqq v_k$, where v_k is assumed to satisfy (5.27), (5.32) and (5.33). Condition (3.13) in Csörgő and Lin (1989) implies (3.20). Thus Theorem 3.3 implies Theorem 3.2 in the latter paper.

Remark 3.5. We note that the statements of (3.5), (3.6), as well as those of (3.11), (3.12) and (3.18), (3.19) hold true with " $\leqq 1$ a.s." instead of their present " $= 1$ a.s." statements in case the respectively indicated increments preceding these statements were not assumed to have negative or zero correlations.

4. Proofs of lemmata

Proof of Lemma 2.1. Write

$$\begin{aligned} |X(t+s, V) - X(t, V)| &\leqq |X((t+s)_r, V) - X(t_r, V)| \\ &\quad + \sum_{j=0}^{\infty} |X((t+s)_{r+j+1}, V) - X((t+s)_{r+j}, V)| \\ &\quad + \sum_{j=0}^{\infty} |X(t_{r+j+1}, V) - X(t_{r+j}, V)| . \end{aligned} \tag{4.1}$$

By conditions (2.1), (2.2) and normality of $X(\cdot,\cdot)$ we have

$$\begin{aligned} &P\left\{\sup_{|t|\leqq T_V} \sup_{0\leqq s\leqq s_V} (H_1(t, s_V, V))^{-1/2}|X((t+s)_r, V) - X(t_r, V)| \geqq \left(1 - \frac{\varepsilon}{6}\right)x\right\} \\ &\leqq P\left\{\sup_{|t|\leqq T_V} \sup_{0\leqq s\leqq s_V} (H_1(t_r, (t+s)_r - t_r, V))^{-1/2}|X((t+s)_r, V) - X(t_r, V)| \geqq x\Big/\left(1 + \frac{\varepsilon}{5}\right)\right\} \\ &\leqq \frac{2T_V s_V}{r'^2} \exp\left\{-\frac{x^2}{2+\varepsilon}\right\} \leqq Cd_V^2 T_V s_V \exp\left\{-\frac{x^2}{2+\varepsilon}\right\}, \end{aligned} \tag{4.2}$$

provided $r = r(\varepsilon)$ is large enough. Furthermore, by condition (2.3)

$$P\left\{\bigcup_{|t|\leqq T_V}\bigcup_{0\leqq s\leqq s_V}\left\{\sum_{j=0}^{\infty}|X((t+s)_{r+j+1},V)-X((t+s)_{r+j},V)|\right.\right.$$
$$\left.\left.\geqq\sum_{j=0}^{\infty}((x^2+g(j))H_1((t+s)_{r+j},r'(r+j+1),V))^{1/2}\right\}\right\}$$
$$\leqq\sum_{j=0}^{\infty}P\left\{\bigcup_{|t|\leqq T_V}\bigcup_{0\leqq s\leqq s_V}\{|X((t+s)_{r+j+1},V)-X((t+s)_{r+j},V)|\right.$$
$$\left.\geqq((x^2+g(j))H_1((t+s)_{r+j},r'(r+j+1),V))^{1/2}\right\}\right\}$$
$$\leqq\sum_{j=0}^{\infty}2e^2(r+j+1)d_V^2T_Vs_V\exp\left\{-\frac{x^2+g(j)}{2+\varepsilon}\right\}$$
$$\leqq Cd_V^2T_Vs_V\exp\left\{-\frac{x^2}{2+\varepsilon}\right\}.$$

We can assume $x\geqq 1$ without loss of generality. Using condition (2.4), we have

$$\sum_{j=0}^{\infty}((x^2+g(j))H_1((t+s)_{r+j},r'(r+j+1),V)^{1/2}\leqq\frac{\varepsilon}{12}x(H_1(t,s_V,V))^{1/2}$$

if r is large enough. Hence

$$P\left\{\sup_{|t|\leqq T_V}\sup_{0\leqq s\leqq s_V}(H_1(t,s_V,V))^{-1/2}\sum_{j=0}^{\infty}|X((t+s)_{r+j+1},V)\right.$$
$$\left.-X((t+s)_{r+j},V)|\geqq\frac{\varepsilon}{12}x\right\}\tag{4.3}$$
$$\leqq Cd_V^2T_Vs_V\exp\left\{-\frac{x^2}{2+\varepsilon}\right\}.$$

Similarly,

$$P\left\{\sup_{|t|\leqq T_V}(H_1(t,s_V,V))^{-1/2}\sum_{j=0}^{\infty}|X(t_{r+j+1},V)-X(t_{r+j},V)|\geqq\frac{\varepsilon}{12}x\right\}\tag{4.4}$$
$$\leqq Cd_V^2T_Vs_V\exp\left\{-\frac{x^2}{2+\varepsilon}\right\}.$$

Combining now (4.1)–(4.4), we get (2.5).

The lines of the proof of (2.8) are similar. As an example, we rewrite here (4.2) in the latter context. By (2.1) and (2.6) we have

$$P\left\{\sup_{V_1\leqq v\leqq V}\sup_{|t|\leqq T_V}\sup_{0\leqq s\leqq s_V}(H_1(t,s_V,v))^{-1/2}|X((t+s)_r,v)-X(t_r,v)|\right.\tag{4.5}$$
$$\left.\geqq\left(1-\frac{\varepsilon}{6}\right)x\right\}$$
$$\leqq\frac{2T_Vs_V}{r'^2}\sup_{|t|\leqq T_V}\sup_{0\leqq s\leqq s_V}P\left\{\sup_{V_1\leqq v\leqq V}|X((t+s)_r,v)-X(t_r,v)|\right.$$
$$\left.\geqq x(H_1(t_r,(t+s)_r-t_r,V))^{1/2}\Big/\left(1+\frac{\varepsilon}{5}\right)\right\}$$

and with $H_n = \left\{V_1 + \frac{i}{n}(V - V_1),\ i = 0, 1, \ldots, n\right\}$ we can write

(4.6)

$$P\left\{\sup_{V_1 \leqq v \leqq V} |X((t+s)_r, v) - X(t_r, v)| \geqq x(H_1(t_r,(t+s)_r - t_r, V))^{1/2} \Big/ \left(1 + \frac{\varepsilon}{5}\right)\right\}$$

$$:= P\left\{\sup_{V_1 \leqq v \leqq V} |Y(v)| \geqq y\right\} = \lim_{n\to\infty} P\left\{\max_{i \in H_n} |Y(v_i)| \geqq y\right\}$$

$$\leqq \lim_{n\to\infty}\left(P\left\{\bigcup_{i\in H_n} (Y(v_i) \geqq y)\right\} + \left(P\left\{\bigcup_{i\in H_n} (Y(v_i) \leqq -y)\right\}\right)\right)$$

$$= P\left\{\sup_{V_1 \leqq v \leqq V} Y(v) \geqq y\right\} + P\left\{\inf_{V_1 \leqq v \leqq V} Y(v) \leqq -y\right\}$$

$$= 2 - \left(P\left\{\sup_{V_1 \leqq v \leqq V} Y(v) < y\right\} + P\left\{\inf_{V_1 \leqq v \leqq V} Y(v) > -y\right\}\right).$$

Condition (2.7) implies that the increments of the process $\{X(t+s, v) - X(t, v),\ 0 \leqq v \leqq V\}$ are non-negatively correlated. Hence if we let $\{Z(v)\}$ be a process with independent increments and such that $Z(v) \overset{\mathcal{D}}{=} Y(v)$ for every v, then by Lemma 2.5 the right hand side of the inequality of (4.6) does not exceed

$$2 - \left(P\left\{\sup_{V_1 \leqq v \leqq V} Z(v) < y\right\} + P\left\{\inf_{V_1 \leqq v \leqq V} Z(v) > -y\right\}\right)$$

$$= P\left\{\sup_{V_1 \leqq v \leqq V} Z(v) \geqq y\right\} + P\left\{\inf_{V_1 \leqq v \leqq V} Z(v) \leqq -y\right\}$$

$$\leqq 2P\left\{\sup_{V_1 \leqq v \leqq V} |Z(v)| \geqq y\right\}.$$

Hence the latter combined with (4.6) and (4.5) results in saying that the right hand side of the inequality (4.5) does not exceed

$$\frac{4T_V s_V}{r'^2} \exp\left\{-\frac{x^2}{2+\varepsilon}\right\} \leqq C d_V^2 T_V s_V \exp\left\{-\frac{x^2}{2+\varepsilon}\right\}. \tag{4.7}$$

Proof of Lemma 2.2. We put $r^* = [u_V/2^r]$, $u^{r^*} = [u/r^*]r^*$ and write

$$\begin{aligned}
(4.8)\quad |X(R(t,s,v,u))| \leqq\ & |X(R(t_r, (t+s)_r - t_r, v^{r^*}, (v+u)^{r^*} - v^{r^*}))| \\
&+ \sum_{j=0}^{\infty} |X(R((t+s)_{r+j}, r'(r+j+1), v^{r^*}, (v+u)^{r^*} - v^{r^*}))| \\
&+ \sum_{j=0}^{\infty} |X(R(t_{r+j}, r'(r+j+1), v^{r^*}, (v+u)^{r^*} - v^{r^*}))| \\
&+ |X(R(t, s, v^{r^*}, v - v^{r^*}))| \\
&+ |X(R(t, s, (v+u)^{r^*}, v + u - (v+u)^{r^*}))|.
\end{aligned}$$

By condition (2.9),

$$H_3(t_r, (t+s)_r - t_r, v^{r^*}, (v+u)^{r^*} - v^{r^*})/H_3(t, s_V, v, u_V) \leqq 1 + \frac{\varepsilon}{30}$$

provided that $r = r(\varepsilon)$ is large enough. Then

$$(4.9)\quad P\Big\{\sup_{0\le v\le V}\ \sup_{0\le u\le u_V}\ \sup_{|t|\le T_V}\ \sup_{0\le s\le s_V} (H_3(t, s_V, v, u_V))^{-1/2} \times |X(R(t_r, (t+s)_r - t_r, v^{r^*}, (v+u)^{r^*} - v^{r^*}))| \ge x\Big\}$$
$$\le 2\frac{Vu_V}{r^{*2}}\frac{T_V s_V}{r'^2}\exp\left\{-\frac{x^2}{2+\varepsilon}\right\} \le C\frac{d_V^2 VT_V s_V}{u_V}\exp\left\{-\frac{x^2}{2+\varepsilon}\right\}.$$

Using condition (2.11) instead of (2.4), the second and third terms on the right hand side of the inequality (4.8) can be treated like those on the right hand side of the inequality of (4.1). Regarding the fourth term on the right hand side of the inequality (4.8), we proceed along the lines of the proof of (2.8) as in (4.5)–(4.7), and using (2.9), (2.10) and Lemma 2.5 with condition (2.7) we obtain

$$(4.10)\quad P\Big\{\sup_{0\le v\le V}\ \sup_{|t|\le T_V}\ \sup_{0\le s\le s_V} (H_3(t, s_V, v, u_V))^{-1/2}|X(R(t, s, v^{r^*}, v - v^{r^*}))| \ge \frac{\varepsilon}{24}x\Big\}$$
$$\le P\Big\{\max_{0\le j\le [V/r^*]}\ \sup_{|t|\le T_V}\ \sup_{0\le s\le s_V}\ \sup_{0\le w\le r^*}\left(\left(\frac{24}{\varepsilon}\right)^2 H_3(t, s_V, jr^*, r^*)\right)^{-1/2} \times |X(R(t, s, jr^*, w)| \ge \frac{\varepsilon}{24}x\Big\}$$
$$\le \frac{Cd_V^2 VT_V s_V}{u_V}\exp\left\{-\frac{x^2}{2+\varepsilon}\right\},$$

provided that r is large enough. The same conclusion holds true for the fifth term on the right hand side of the inequality (4.8), and (2.12) now follows from combining these inequalities.

Proof of Lemma 2.3. We have

$$(4.11)\quad |X(t+s, v) - X(t, v)| \le |X((t+s)_r, v^r) - X(t_r, v^r)|$$
$$+ \sum_{j=0}^{\infty} |X((t+s)_{r+j+1}, v^r) - X((t+s)_{r+j}, v^r)|$$
$$+ \sum_{j=0}^{\infty} |X(t_{r+j+1}, v^r) - X(t_{r+j}, v^r)|$$
$$+ \sum_{j=0}^{\infty} |X((t+s)_r, v^{r+j+1}) - X((t+s)_r, v^{r+j})|$$
$$+ \sum_{j=0}^{\infty} |X(t_r, v^{r+j+1}) - X(t_r, v^{r+j})|$$
$$+ \sum_{i=0}^{\infty}\sum_{j=0}^{\infty} |X(R((t+s)_{r+i}, r'(r+i+1), v^{r+j}, r''(r+j+1)))|$$
$$+ \sum_{i=0}^{\infty}\sum_{j=0}^{\infty} |X(R(t_{r+i}, r'(r+i+1), v^{r+j}, r''(r+j+1)))|.$$

By condition (2.13)

$$P\left\{\sup_{0\leqq v\leqq V}\sup_{|t|\leqq T_V}\sup_{0\leqq s\leqq s_V}(H_1(t,s_V,v))^{-1/2}|X((t+s)_r,v^r)-X(t_r,v^r)|\geqq\left(1-\frac{\varepsilon}{6}x\right)\right\}$$
$$\leqq Cd'_V d_V^2 VT_V s_V \exp\left\{-\frac{x^2}{2+\varepsilon}\right\}.$$

For the second and third parts of the right hand side of the inequality (4.11), by using (2.13), (2.2) and (2.14), and imitating the proof of (4.3), we obtain

$$P\left\{\sup_{0\leqq v\leqq V}\sup_{|t|\leqq T_V}\sup_{0\leqq s\leqq s_V}(H_1(t,s_V,v))^{-1/2}\sum_{j=0}^{\infty}|X((t+s)_{r+j+1},v^r)-X(t_{r+j},v^r)|\right.$$
$$\left.\geqq\frac{\varepsilon}{36}x\right\}\leqq Cd'_V d_V^2 VT_V s_V\exp\left\{-\frac{x^2}{2+\varepsilon}\right\}.$$

The same method is applied to deal with the rest of the right hand side of the inequality (4.11), and hence we omit these details which, when combined with the above inequalities, also yield (2.19).

5. Proofs of Theorems

Proof of Theorem 3.1. For a given $\varepsilon>0$ let v_k, $k=1,2,\ldots,$ be an increasing sequence such that

$$\sum_{k=1}^{\infty}(T_{v_k}/s_{v_{k-1}})^{-\varepsilon/2}(\log v_k)^{-(1+\varepsilon)}<\infty\ , \tag{5.1}$$

$$(T_{v_{k+1}}/s_{v_k})/(T_{v_k}/s_{v_{k-1}})\to 1\quad\text{as } k\to\infty \tag{5.2}$$

and $(v_k-v_{k-1})/v_k\to 0$ as $k\to\infty$.

Hence, by (2.8) and (3.1), we have

$$\begin{aligned}(5.3)\qquad & P\left\{\sup_{v_{k-1}\leqq v\leqq v_k}\sup_{|t|\leqq T_{v_k}}\sup_{0\leqq s\leqq s_{v_{k-1}}}(2H_1(t,s_{v_{k-1}},v)\log(T_{v_k}/s_{v_{k-1}}))^{-1/2}\right.\\ &\left.\times|X(t+s,v)-X(t,v)|\geqq 1+\frac{4}{5}\varepsilon\right\}\\ &\leqslant Cd_{v_k}^2T_{v_k}s_{v_{k-1}}\exp\left\{-\frac{2(1+4\varepsilon/5)^2}{2+\varepsilon}\log(T_{v_k}s_{v_{k-1}}^{-1}\log v_k)\right\}\\ &\leqq C(T_{v_k}s_{v_{k-1}}^{-1})^{1+\varepsilon/2}(T_{v_k}s_{v_{k-1}}^{-1}\log v_k)^{-(1+\varepsilon)}\\ &\leqq C(T_{v_k}s_{v_{k-1}}^{-1})^{-\varepsilon/2}(\log v_k)^{-(1+\varepsilon)}.\end{aligned}$$

By (5.1)

$$\limsup_{k\to\infty}\sup_{v_{k-1}\leqq v\leqq v_k}\sup_{|t|\leqq T_{v_k}}\sup_{0\leqq s\leqq s_{v_{k-1}}}(2H_1(t,s_{v_{k-1}},v)\log(T_{v_k}/s_{v_{k-1}}))^{-1/2}$$
$$\times|X(t+s,v)-X(t,v)|\leqq 1+\frac{4}{5}\varepsilon\quad\text{a.s.}$$

and by (2.1), (5.2)

$$(5.4)\quad \limsup_{k\to\infty} \sup_{v_{k-1}\leqq v\leqq v_k} \sup_{|t|\leqq T_{v_k}} \sup_{0\leqq s\leqq s_{v_{k-1}}} (2H_1(t, s_v, v)\log(T_v/s_v))^{-1/2} \times |X(t+s, v) - X(t, v)| \leqq 1+\varepsilon \quad \text{a.s.}$$

Hence

$$(5.5)\quad \limsup_{V\to\infty} \sup_{|t|\leqq T_V} \sup_{0\leqq s\leqq s_V} (2H_1(t, s_V, V)\log(T_V/s_V))^{-1/2} \times |X(t+s, V) - X(t, V)| \leqq 1+\varepsilon \quad \text{a.s.}$$

Next we prove

$$(5.6)\quad \liminf_{V\to\infty} \sup_{|t|\leqq T_V} (2H_1(t, s_V, V)\log(T_V/s_V))^{-1/2}|X(t+s_v, V) - X(t, V)| \geqq 1-\varepsilon \quad \text{a.s.}$$

via showing

$$(5.7)\quad \liminf_{V\to\infty} \sup_{|t|\leqq T_V} (2H_1(t, s_V, V)\log(T_V/s_V))^{-1/2}(X(t+s_v, V) - X(t, V)) \geqq 1-\varepsilon \quad \text{a.s.}$$

Given that the increments $\{X((k+1)s_V, V) - X(ks_V, V),\ |k| \leqq [T_V/s_V]\}$ are assumed to be not positively correlated, by Lemma 2.5 we obtain

$$(5.8)\quad P\left\{\max_{|k|\leqq[T_V/s_V]} (2H_1(ks_V, s_V, V)\log(T_V/s_V))^{-1/2}(X((k+1)s_V, V) - X(ks_V, V)) \leqq 1-\varepsilon\right\}$$

$$\leqq \left(P\{(2H_1(ks_V, s_V, V)\log(T_V/s_V))^{-1/2}(X((k+1)s_V, V) - X(ks_V, V)) \leqq 1-\varepsilon\}\right)^{2[T_V/s_V]}$$

$$\leqq \left(1 - \frac{1}{6(\log(T_V/s_V))^{1/2}} \exp\{-(1-\varepsilon)^2\log(T_V/s_V)\}\right)^{2[T_V/s_V]}$$

$$\leqq \exp\{-(T_V/s_V)^{\varepsilon}\} \leqq \exp\{-(T_V/s_V)^{\varepsilon/2}\log V\},$$

where the last inequality is due to (3.1). Let v_k be a sequence satisfying (5.2) as well as

$$(5.9)\quad \sum_{k=1}^{\infty} \exp\{-(T_{v_k}/s_{v_k})^{\varepsilon/2}\log v_k\} < \infty,$$

$$(5.10)\quad \limsup_{k\to\infty} \frac{\log(s_{v_{k-1}} - s_{v_k})^{-1}}{\log(T_{v_k}/s_{v_k})} < \infty.$$

Hence (5.8) implies

$$(5.11)\quad \liminf_{k\to\infty} \sup_{|t|\leqq T_{v_k}} (2H_1(t, s_{v_k}, v_k)\log(T_{v_k}/s_{v_k}))^{-1/2}|X(t+s_{v_k}, v_k) - X(t, v_k)| \geqq 1-\varepsilon \quad \text{a.s.}$$

We note that Lemma 2.1 is applicable with $s_{v_{k-1}} - s_{v_k}$ instead of s_{v_k} if d_V does not depend on s_V. Hence, imitating the proof of (5.4) via (5.3) and noting that $s_{v_{k-1}} - s_{v_k} \leqq s_{v_k}$ (recalling the second inequality of (5.3)), we get

$$\limsup_{k\to\infty} \sup_{v_{k-1}\leqq v\leqq v_k} \sup_{|t|\leqq T_{v_k}} (2H_1(t+s_{v_k}, s_{v_{k-1}} - s_{v_k}, v)\log(T_{v_k}/s_{v_k}))^{-1/2} \times |X(t+s_v, v) - X(t+s_{v_k}, v)| \leqq 1+\varepsilon \quad \text{a.s.}$$

which, in turn, implies

$$\limsup_{k\to\infty} \sup_{v_{k-1}\leqq v\leqq v_k} \sup_{|t|\leqq T_{v_k}} (2H_1(t, s_v, v)\log(T_v/s_v))^{-1/2} \times |X(t+s_v, v) - X(t+s_{v_k}, v)| = 0 \quad \text{a.s.} \tag{5.12}$$

by using conditions (3.3) and (5.2). If condition (3.4) is satisfied, then we have

$$\limsup_{k\to\infty} \sup_{v_{k-1}\leqq v\leqq v_k} \sup_{|t|\leqq T_{v_k}} (2H_1(t+s_{v_k}, s_{v_{k-1}} - s_{v_k}, v)\log(T_{v_k}(s_{v_{k-1}} - s_{v_k})^{-1}))^{-1/2} \times |X(t+s_v, v) - X(t+s_{v_k}, v)| \leqq 1+\varepsilon \quad \text{a.s.}$$

by taking $d_{v_k} = (s_{v_{k-1}} - s_{v_k})^{-1}$ (recalling condition (2.4)), which again implies (5.12) with the help of (3.4) and (5.10).

By using conditions (3.1), (2.7), (2.11), (2.1) and (3.2) (noting that we can take $v_k/k^\delta \to 0$, as $k\to\infty$, for any $\delta > 0$), imitating the proof of Lemma 2.2 and then applying the Borel–Cantelli lemma, we can prove

$$\limsup_{k\to\infty} \sup_{v_{k-1}\leqq v\leqq v_k} \sup_{|t|\leqq T_{v_k}} (2H_1(t, s_{v_k}, v_k)\log(T_{v_k}/s_{v_k}))^{-1/2} \times |X(R(t, s_{v_k}, v, v_k - v))| = 0 \quad \text{a.s.}$$

which, in turn, implies

$$\limsup_{k\to\infty} \sup_{v_{k-1}\leqq v\leqq v_k} \sup_{|t|\leqq T_{v_k}} (2H_1(t, s_v, v)\log(T_v/s_v))^{-1/2} \times |X(R(t, s_{v_k}, v, v_k - v))| = 0 \quad \text{a.s.}, \tag{5.13}$$

on noting conditions (2.1), (2.6) and (5.2). Hence, combining (5.11), (5.12) and (5.13) we get (5.6), and (5.5), (5.6) imply the conclusion of Theorem 3.1.

Proof of Theorem 3.2. First we assume that $u_V \downarrow 0$ as $V\to\infty$. The proof is similar when u_V is increasing. Let $\{v_k\}$ be a sequence such that

$$\sum_{k=1}^{\infty} (v_k T_{v_k} s_{v_k}^{-1} u_{v_k}^{-1})^{-\varepsilon/2} < \infty, \tag{5.14}$$

$$\frac{v_{k+1} T_{v_{k+1}} s_{v_k}^{-1} u_{v_k}^{-1}}{v_k T_{v_k} s_{v_{k-1}}^{-1} u_{v_{k-1}}^{-1}} \to 1 \quad \text{as} \quad k\to\infty. \tag{5.15}$$

From (2.12) and (3.10) we have

$$P\left\{\sup_{0\leqq v\leqq v_k} \sup_{0\leqq u\leqq u_{v_{k-1}}} \sup_{|t|\leqq T_{v_k}} \sup_{0\leqq s\leqq s_{v_{k-1}}} (2H_3(t, s_{v_{k-1}}, v, u_{v_{k-1}}) \times \log(v_k T_{v_k} s_{v_{k-1}}^{-1} u_{v_{k-1}}^{-1}))^{-1/2} |X(R(t,s,v,u))| \geqq 1 + \frac{4}{5}\varepsilon\right\} \tag{5.16}$$

$$\leqq C d_{v_k}^2 v_k T_{v_k} s_{v_{k-1}} u_{v_{k-1}}^{-1} \exp\left\{-2\left(\left(1+\frac{4}{5}\varepsilon\right)^2 \Big/ (2+\varepsilon)\right)\log(v_k T_{v_k} s_{v_{k-1}}^{-1} u_{v_{k-1}}^{-1})\right\}$$

$$\leqq C(v_k T_{v_k} s_{v_{k-1}}^{-1} u_{v_{k-1}}^{-1})^{1+\varepsilon/2}(v_k T_{v_k} s_{v_{k-1}}^{-1} u_{v_{k-1}}^{-1})^{-(1+\varepsilon)}$$
$$= C(v_k T_{v_k} s_{v_{k-1}}^{-1} u_{v_{k-1}}^{-1})^{-\varepsilon/2} .$$

Therefore, by using (5.14), we obtain

$$\limsup_{k\to\infty} \sup_{0\leqq v\leqq v_k} \sup_{0\leqq u\leqq u_{v_{k-1}}} \sup_{|t|\leqq T_{v_k}} \sup_{0\leqq s\leqq s_{v_{k-1}}} (2H_3(t, s_{v_{k-1}}, v, u_{v_{k-1}})$$
$$\times \log(v_k T_{v_k} s_{v_{k-1}}^{-1} u_{v_{k-1}}^{-1}))^{-1/2}|X(R(t, s, v, u))| \leqq 1 + \frac{4}{5}\varepsilon \quad \text{a.s.}$$

which, with the help of (2.9) and (5.15), implies

$$\limsup_{V\to\infty} \sup_{0\leqq v\leqq V} \sup_{0\leqq u\leqq u_V} \sup_{|t|\leqq T_V} \sup_{0\leqq s\leqq s_V} (2H_3(t, s_V, v, u_V) \tag{5.17}$$
$$\times \log(VT_V s_V^{-1} u_V^{-1}))^{-1/2}|X(R(t, s, v, u))| \leqq 1 + \varepsilon \quad \text{a.s.}$$

Next we prove

$$\liminf_{V\to\infty} \sup_{0\leqq v\leqq V} \sup_{|t|\leqq T_V} (2H_3(t, s_V, v, u_V)\log(VT_V s_V^{-1} u_V^{-1}))^{-1/2} \tag{5.18}$$
$$\times |X(R(t, s_V, v, u_V))| \geqq 1 - \varepsilon \quad \text{a.s.}$$

Because the increments $\{X(R(ks_V, s_V, lu_V, u_V))\}$ in k as well as in l are assumed to be not positively correlated, by Lemma 2.5 we get

$$P\left\{\max_{0\leqq l\leqq [Vu_V^{-1}]} \max_{|k|\leqq [T_V s_V^{-1}]} (2H_3(ks_V, s_V, lu_V, u_V)\log(VT_V s_V^{-1} u_V^{-1}))^{-1/2}\right. \tag{5.19}$$
$$\left.\times (X(R(ks_V, s_V, lu_V, u_V))) \leqq 1 - \varepsilon\right\}$$
$$\leqq \prod_{l=0}^{[Vu_V^{-1}]} \sum_{k=-[T_V s_V^{-1}]}^{[T_V s_V^{-1}]} P\left\{(2H_3(ks_V, s_V, lu_V, u_V)\log(VT_V s_V^{-1} u_V^{-1}))^{-1/2}\right.$$
$$\left.\times (X(R(ks_V, s_V, lu_V, u_V))) \leqq 1 - \varepsilon\right\}$$
$$\leqq \left\{1 - \frac{1}{6(\log(VT_V s_V^{-1} u_V^{-1}))^{1/2}} \exp\left\{-(1-\varepsilon)^2\right.\right.$$
$$\left.\left.\times \log(VT_V s_V^{-1} u_V^{-1})\right\}\right\}^{2VT_V s_V^{-1} u_V^{-1}}$$
$$\leqq \exp\{-(VT_V s_V^{-1} u_V^{-1})^{\varepsilon}\}$$

Let $\{v_k\}$ be a sequence satisfying (5.15), as well as

$$\sum_{k=1}^{\infty} \exp\{-(v_k T_{v_k} s_{v_k}^{-1} u_{v_k}^{-1})^{\varepsilon}\} < \infty \tag{5.20}$$

and

$$\limsup_{k\to\infty} \frac{\log(u_{v_{k-1}} - u_{v_k})^{-1}}{\log(v_k T_{v_k} s_{v_k}^{-1} u_{v_k}^{-1})} < \infty , \quad \limsup_{k\to\infty} \frac{\log(s_{v_{k-1}} - s_{v_k})^{-1}}{\log(v_k T_{v_k} s_{v_k}^{-1} u_{v_k}^{-1})} < \infty . \tag{5.21}$$

Then we have

$$\liminf_{k\to\infty} \sup_{0\leqq v\leqq v_k} \sup_{|t|\leqq T_{v_k}} (2H_3(t, s_{v_k}, v, u_{v_k}) \log(v_k T_{v_k} s_{v_k}^{-1} u_{v_k}^{-1}))^{-1/2} \times |X(R(t, s_{v_k}, v, u_{v_k}))| \geqq 1 - \varepsilon \quad \text{a.s.} \tag{5.22}$$

Similarly to the proof of (5.17), we can prove also

$$\limsup_{k\to\infty} \sup_{0\leqq v\leqq v_k} \sup_{0\leqq u\leqq u_{v_{k-1}}-u_{v_k}} \sup_{|t|\leqq T_{v_k}} (2H_3(t, s_{v_k}, v+u_{v_k}, u_{v_{k-1}}-u_{v_k}) \times \log(v_k T_{v_k} s_{v_k}^{-1}(u_{v_{k-1}}-u_{v_k})^{-1}))^{-1/2} |X(R(t, s_{v_k}, v+u_{v_k}, u))| \leqq 1+\varepsilon \quad \text{a.s.}$$

Since we can also apply Lemma 2.2 when u_{v_k} is replaced by $u_{v_{k-1}} - u_{v_k}$, hence by using conditions (2.9), (2.10) and (5.21) we obtain

$$\limsup_{k\to\infty} \sup_{0\leqq v\leqq v_k} \sup_{0\leqq u\leqq u_{v_{k-1}}-u_{v_k}} \sup_{|t|\leqq T_{v_k}} (2H_3(t, s_{v_k}, v, u_{v_k}) \times \log(v_k T_{v_k} s_{v_k}^{-1} u_{v_k}^{-1}))^{-1/2} |X(R(t, s_{v_k}, v+u_{v_k}, u))| = 0 \quad \text{a.s.} \tag{5.23}$$

We can again apply Lemma 2.2, this time with $s_{v_{k-1}} - s_{v_k}$ if d_V does not depend on s_V. Hence, imitating the proof of (5.17) and noting that $s_{v_{k-1}} - s_{v_k} \leqq s_{v_k}$ (recalling the second inequality of (5.16)), we have

$$\limsup_{k\to\infty} \sup_{0\leqq v\leqq v_k} \sup_{|t|\leqq T_{v_k}} \sup_{0\leqq s\leqq s_{v_{k-1}}-s_{v_k}} (2H_3(t+s_{v_k}, s_{v_{k-1}}-s_{v_k}, v, u_{v_k}) \times \log(v_k T_{v_k} s_{v_k}^{-1} u_{v_k}^{-1}))^{-1/2} |X(R(t+s_{v_k}, s, v, u_{v_k}))| \leqq 1+\varepsilon \quad \text{a.s.}$$

and this, on using conditions (2.9) and (3.8), yields

$$\limsup_{k\to\infty} \sup_{0\leqq v\leqq v_k} \sup_{|t|\leqq T_{v_k}} \sup_{0\leqq s\leqq s_{v_{k-1}}-s_{v_k}} (2H_3(t, s_{v_k}, v, u_{v_k}) \times \log(v_k T_{v_k} s_{v_k}^{-1} u_{v_k}^{-1}))^{-1/2} |X(R(t+s_{v_k}, s, v, u_{v_k}))| = 0 \quad \text{a.s.} \tag{5.24}$$

Also, if condition (3.9) is satisfied, then by taking $d_{v_k} = (s_{v_{k-1}} - s_{v_k})^{-1}$ and recalling condition (2.11) we get

$$\limsup_{k\to\infty} \sup_{0\leqq v\leqq v_k} \sup_{|t|\leqq T_{v_k}} \sup_{0\leqq s\leqq s_{v_{k-1}}-s_{v_k}} (2H_3(t+s_{v_k}, s_{v_{k-1}}-s_{v_k}, v, u_{v_k}) \times \log(v_k T_{v_k}(s_{v_{k-1}}-s_{v_k})^{-1} u_{v_k}^{-1}))^{-1/2} |X(R(t+s_{v_k}, s, v, u_{v_k}))| \leqq 1+\varepsilon \quad \text{a.s.}$$

This again implies (5.24) with the help of (2.9), (3.9) and (5.21). Similarly,

$$\limsup_{k\to\infty} \sup_{0\leqq v\leqq v_k} \sup_{0\leqq u\leqq u_{v_{k-1}}-u_{v_k}} \sup_{|t|\leqq T_{v_k}} \sup_{0\leqq s\leqq s_{v_{k-1}}-s_{v_k}} (2H_3(t, s_{v_k}, v, u_{v_k}) \times \log(v_k T_{v_k} s_{v_k}^{-1} u_{v_k}^{-1}))^{-1/2} |X(R(t+s_{v_k}, s, v+u_{v_k}, u))| = 0 \quad \text{a.s.} \tag{5.25}$$

Combining (5.22)–(5.25) we get (5.18). The latter and (5.17) imply the conclusions of Theorem 3.2.

Proof of Theorem 3.3. Choose $\{v_k\}$ such that

$$\sum_{k=1}^{\infty} (v_k u_{v_{k-1}}^{-1})^{-\varepsilon/2} < \infty \tag{5.26}$$

and

$$(v_{k+1} u_{v_k}^{-1})/(v_k u_{v_{k-1}}^{-1}) \to 1 \quad \text{as} \quad k \to \infty\,. \tag{5.27}$$

By Lemma 2.4 and condition (3.14)

$$P\left\{\sup_{0\leqq v\leqq v_k}\sup_{0\leqq u\leqq u_{v_{k-1}}}\sup_{|t|\leqq T_{v_k}}(2H_2(t,v,u_{v_{k-1}})\log(v_k u_{v_{k-1}}^{-1}))^{-1/2}\right.$$
$$\left.\times|X(t,v+u)-X(t,v)|\geqq 1+\frac{4}{5}\varepsilon\right\}$$
$$\leqq Cd_{v_k}d_{v_k}'^2 v_k T_{v_k}u_{v_{k-1}}\exp\left\{-\frac{2(1+4\varepsilon/5)^2}{2+\varepsilon}\log(v_k u_{v_{k-1}}^{-1})\right\}$$
$$\leqq C(v_k u_{v_{k-1}}^{-1})^{1+\varepsilon/2}(v_k u_{v_{k-1}}^{-1})^{-(1+\varepsilon)}\leqq C(v_k u_{v_{k-1}}^{-1})^{-\varepsilon/2}$$

and hence by (5.26) we have

$$\limsup_{k\to\infty}\sup_{0\leqq v\leqq v_k}\sup_{0\leqq u\leqq u_{v_{k-1}}}\sup_{|t|\leqq T_{v_k}}(2H_2(t,v,u_{v_{k-1}})\log(v_k u_{v_{k-1}}^{-1}))^{-1/2}$$
$$\times|X(t,v+u)-X(t,v)|\leqq 1+\frac{4}{5}\varepsilon\quad\text{a.s.}\tag{5.28}$$

Consequently, by (2.20) and (5.27) we obtain

$$\limsup_{V\to\infty}\sup_{0\leqq v\leqq V}\sup_{0\leqq u\leqq u_V}\sup_{|t|\leqq T_V}(2H_2(t,v,u_V)\log(Vu_V^{-1}))^{-1/2}$$
$$\times|X(t,v+u)-X(t,v)|\leqq 1+\varepsilon\quad\text{a.s.}\tag{5.29}$$

Next we prove

$$\liminf_{V\to\infty}\sup_{0\leqq v\leqq V}(2H_2(T_V,v,u_V)\log(Vu_V^{-1}))^{-1/2}|X(T_V,v+u_V)-X(T_V,v)|$$
$$\geqq 1-\varepsilon\quad\text{a.s.}\tag{5.30}$$

The assumption that the increments $\{X(T_V,(k+1)u_V)-X(T_V,ku_V),\ 0\leqq k\leqq[V/u_V]\}$ are not positively correlated and Lemma 2.5 yield

$$P\left\{\max_{0\leqq k\leqq[Vu_V^{-1}]}(2H_2(T_V,ku_V,u_V)\log(Vu_V^{-1}))^{-1/2}\right.$$
$$\left.\times(X(T_V,(k+1)u_V)-X(T_V,ku_V))\leqq 1-\varepsilon\right\}\tag{5.31}$$
$$\leqq\prod_{k=0}^{[Vu_V^{-1}]}P\{(2H_2(T_V,ku_V,u_V)\log(Vu_V^{-1}))^{-1/2}$$
$$\times(X(T_V,(k+1)u_V)-X(T_V,ku_V))\leqq 1-\varepsilon\}$$
$$\leqq\left\{1-\frac{1}{6(\log(Vu_V^{-1}))^{1/2}}\exp\{-(1-\varepsilon)^2\log(Vu_V^{-1})\}\right\}^{[Vu_V^{-1}]}$$
$$\leqq\exp\{-Vu_V^{-1})^{\varepsilon}\}\,.$$

Let $\{v_k\}$ be a sequence satisfying (5.27) and

$$\sum_{k=1}^{\infty}\exp\{-(v_k u_{v_k}^{-1})^{\varepsilon}\}<\infty\,,\tag{5.32}$$

$$\lim_{k\to\infty}\frac{\log(u_{v_{k-1}}-u_{v_k})^{-1}}{\log(v_k u_{v_k}^{-1})}<\infty\,.\tag{5.33}$$

Then, by (5.31)

$$\liminf_{k\to\infty} \sup_{0\leqq v\leqq v_k} (2H_2(T_{v_k}, v, u_{v_k})\log(v_k u_{v_k}^{-1}))^{-1/2} \times |X(T_{v_k}, v+u_{v_k}) - X(T_{v_k}, v)| \geqq 1-\varepsilon \quad \text{a.s.} \tag{5.34}$$

We can apply Lemma 2.4 with $u_{v_{k-1}} - u_{v_k}$ instead of u_{v_k} if d'_V does not depend on u_V. Hence, imitating the proof of (5.29) and noting that $u_{v_{k-1}} - u_{v_k} \leqq u_{v_k}$, we obtain

$$\limsup_{k\to\infty} \sup_{0\leqq v\leqq v_k} \sup_{0\leqq u\leqq u_{v_{k-1}}-u_{v_k}} \sup_{T_{v_{k-1}}<t\leqq T_{v_k}} (2H_2(t, v+u_{v_k}, u_{v_{k-1}}-u_{v_k}) \times \log(v_k u_{v_k}^{-1}))^{-1/2} |X(t, v+u_{v_k}+u) - X(t, v+u_{v_k})| \leqq 1+\varepsilon \quad \text{a.s.}$$

which, with the help of (3.16), implies

$$\limsup_{k\to\infty} \sup_{0\leqq v\leqq v_k} \sup_{0\leqq u\leqq u_{v_{k-1}}-u_{v_k}} \sup_{T_{v_{k-1}}<t\leqq T_{v_k}} (2H_2(t, v+u_{v_k}, u_{v_k}) \times \log(v_k u_{v_k}^{-1}))^{-1/2} |X(t, v+u_{v_k}+u) - X(t, v+u_{v_k})| = 0 \quad \text{a.s.} \tag{5.35}$$

By condition (3.17) we have

$$\frac{H_3(t, s, v, \delta u_V)}{H_2(t, v, u_V)} \leqq b'\delta \quad \text{for} \quad s = o(T_V) \tag{5.36}$$

and some $b' > 0$. Using (3.17) and (5.36) we can prove

$$\limsup_{k\to\infty} \sup_{0\leqq v\leqq v_k} \sup_{0\leqq u\leqq u_{v_{k-1}}-u_{v_k}} \sup_{T_{v_{k-1}}<t\leqq T_{v_k}} (2H_2(t, v+u_{v_k}, u_{v_{k-1}}-u_{v_k}) \times \log(v_k(u_{v_{k-1}}-u_{v_k})^{-1}))^{-1/2} |X(t, v+u_{v_k}+u) - X(t, v+u_{v_k})| \leqq 1+\varepsilon \quad \text{a.s.}$$

by taking $d'_{v_k} = (u_{v_{k-1}} - u_{v_k})^{-1}$ and recalling conditions (2.21) and (2.23). This, in turn, again implies (5.35) with the help of (3.17) and (5.33). Moreover, by (5.27) and (3.14), $T_{v_k} \leqq k^{\varepsilon}$ for any $\varepsilon > 0$. Hence, by condition (3.15) and Lemma 2.2 we have

$$\limsup_{k\to\infty} \sup_{0\leqq v\leqq v_k} \sup_{0\leqq s\leqq T_{v_k}-T_{v_{k-1}}} \sup_{T_{v_{k-1}}<t\leqq T_{v_k}} (2H_2(t, v, u_{v_k})\log(v_k u_{v_k}^{-1}))^{-1/2} \times |X(R(t, s, v, u_{v_k}))| = 0 \quad \text{a.s.}$$

Combining (5.34), (5.35) with (3.20) results in (5.30), and the latter with (5.29) also completes the proof of Theorem 3.3.

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